

Figma CUA Benchmark — Tasks 1–10

Model: **claude-sonnet-4-5** | Run date: 2026-05-10 | k = 1 | threshold: final_score ≥ 0.7 | prompt-mode: description | no system prompt | keyboard allowed

pass@1	mean score	mean efficiency ×	mean turns	total cost
3/9 (33%)	0.571	0.698	57	\$20.37

Per-task results

#	Task	Pass	Score	Base	Eff×	Turns / target	Time	Cost	Failure mode
01	Two-story house	FAIL	0.622	0.88	0.70	43 / 30	220s	\$1.30	Polygon roof rotated 180° (apex pointing down). Polygon doesn't sit flush above body rectangle. One body rectangle drawn too narrow (width fraction 0.04). All other rubrics 100%.
02	Sunset stripe band	FAIL	0.386	0.60	0.64	39 / 24	203s	\$1.11	Got 5 stripes correctly stacked at uniform size, but ALL fill colors wrong (color diff 0.35 on layer 1; sunset gradient palette not matched). Color rubric scored 0/0.40.
03	Radial flower with petals	FAIL	0.401	0.78	0.52	104 / 30	657s	\$4.63	All 9 ellipses drawn with distinct colors, but the 8 outer petals not arranged radially: gap-angle deviation 9.8° (tol 15° passed) AND radius ratio 1.44 (need ~1.0) failed. Burned 104 turns trying to reposition.
04	Color hexagon ring	FAIL	0.300	0.50	0.60	42 / 24	227s	\$1.25	6 squares created with 7 distinct colors, but NOT arranged in a ring: gap-angle deviation 22° (tol 10°) and radius spread 61px (tol 22°) both fail. Layout looks like a cluster, not a hexagon.
05	Plus-sign emblem	PASS	1.000	1.00	1.00	7 / 15	42s	\$0.14	—
06	Asterisk burst	FAIL	0.246	0.45	0.55	45 / 24	271s	\$1.42	Created 8 lines, but endpoint cluster too weak — only 6/8 endpoints within 20px of shared center. Stroke color is some yellow but not the target gold (#d9a521).
07	Layered mountain range	PASS	0.778	0.95	0.82	41 / 30	230s	\$1.20	Passed at 0.78. All shape, alignment, color, structure rubrics 100%. Only weak spot: `create_vector` event count = 0 (model used pen but the semantic log didn't record vector creations). Above threshold.
08	Layered water waves	PASS	0.778	0.95	0.82	145 / 40	772s	\$7.93	Passed at 0.78. 2 curved vectors with matching stroke weights, distinct stroke colors — all geometric/color rubrics 100%. Same `create_vector` event-count miss as task 07. Burned 145 turns to get there (target 40), pushing cost to \$7.93 — most expensive single task.
09	12-color swatch grid	skipped							skipped
10	Concentric squares	FAIL	0.630	1.00	0.63	44 / 18	244s	\$1.39	All 4 concentric rectangles drawn correctly with alternating fills and proper nesting — base score 1.00. Failed only on efficiency: 45 turns vs 18 target → eff multiplier 0.63.

Patterns

- **Geometric arrangements are the dominant failure mode.** Radial placements (task 03 orb, task 04 wheel, task 06 star) all failed on positioning — the model gets shape counts and colors right, then can't lay them out in a ring or with matching radii.
- **Colors mismatch when palette is specific.** Task 02 (sunset gradient) had 5 stripes correctly sized and stacked but used wrong colors entirely — color rubric scored 0/0.40.
- **Efficiency is the silent killer.** Task 10 nailed all rubrics (base 1.0) but burned 45 turns vs 18 target → final 0.63, missed pass threshold. Task 01 likewise has 0.885 base but 0.70 efficiency multiplier.
- **Small-detail orientation errors.** Task 01 polygon roof rotated 180° (apex pointing down). Same class of error as not knowing canonical orientation.

- **When tasks are simple and few-shape, performance is excellent.** Task 05 (plus sign, 2 rectangles): 1.000 score in 7 turns. Task 07 (mountain range, pen tool): 0.778 in 41 turns.

- **Long-tail cost risk.** Task 08 passed but burned 145 turns and \$7.93 — half the total spend across all 9 tasks. With no step cap, a single stubborn task can dominate budget. Consider re-introducing a cap (e.g. 80–100) for budget safety.

Generated from 9 attempts (6,404,663 input tokens, 77,345 output tokens, \$20.37 spend). Source: [apps/figma/cua-eval/runs/202605](#). Skipped: task 09 (not run yet).*